Talha Khan

Assistant Professor
Department of Information Sciences and Technologies
College of Computing, Grand Valley State University, USA

khanm4@gvsu.edu
https://talhakhan13.github.io/
Google Scholar

Research Interests: Human-Computer Interaction, Mixed Reality, Applied AI

Professional Experience

- **Assistant Professor** — Aug. 2025 - Current
  College of Computing, Grand Valley State University, USA
  - Director, Immersive Interaction Lab
- **Graduate Research Fellow** — Sep. 2024 - Apr. 2025
  Department of Computer Science, University of Pittsburgh, USA
- **Research Intern, Human-Centered AI** — May. 2024 - Aug. 2024
  Toyota Research Institute, USA
- **Research Intern, Augmented Perception Lab** — May. 2023 - Aug. 2023
  Carnegie Mellon University, USA
  Mentor: Dr. David Lindlbauer
- **Graduate Student Researcher** — May. 2021 - Apr. 2024
  Department of Computer Science, University of Pittsburgh, USA

Education

- **PhD in Computer Science** — Aug. 2020 - Apr. 2025
  University of Pittsburgh, USA
  - Thesis: Towards Integrating Mixed Reality Into Physical Workspaces
  - Advisor: Dr. Jacob Biehl
- **MS in Computer Science** — Sep. 2018 - Jun. 2020
  University of California, Irvine, USA
- **BS in Computer Science** — Aug. 2014 - Jun. 2018
  Lahore University of Management Sciences, Pakistan

Teaching Experience

- **Course Instructor** — Aug. 2025 - Current
  Grand Valley State University, USA
  Courses: *Computer Science 1* (Introduction to Python), *AR/VR Design and Research*
- **Graduate Teaching Assistant** — Sep. 2020 - Apr. 2021
  University of Pittsburgh, USA
  Courses: *Algorithm Implementation, Programming Languages for Web Applications*
- **Graduate Teaching Assistant** — Oct. 2018 - May. 2020
  University of California, Irvine, USA
  Courses: *Intermediate Programming (Python), Data Structures (C++), Introduction to C++*
- **Undergraduate Teaching Assistant** — Sep. 2017 - May. 2018
  Lahore University of Management Sciences, Pakistan
  Courses: *Applied Probability (Graduate-level), Probability*

PUBLICATIONS

Conference Publications & Proceedings

[C.11] CommonGround: Externalizing Mixed Reality Interfaces for Active Bystander Collaboration
Khadija Arif, **Talha Khan**
ACM CSCW 2026

[C.10] ReadAR: Utilizing Mixed Reality for Physical Document Digitization
Liam Adams, **Talha Khan**
ACM IMX 2026

[C.9] Utilizing VLMs for Identifying Child Safety Hazards
Talha Khan, Ahmed Ashraff Butt
ACM CHI 2026

[C.8] MR-MDEs: Exploring the Integration of Mixed Reality into Multi-Display Environments
Griffin Hurt, **Talha Khan**, Nicolás Kass, Anthony Tang, Edward Andrews, Jacob Biehl
PACM HCI (ACM ISS 2025)

[C.7] Shadows of Reality: Enhancing Bystander Awareness of Mixed Reality Interfaces
Talha Khan, Abigail Zimmerman, Edward Andrews, David Lindlbauer, Jacob Biehl
ACM SUI 2025

[C.6] Don't Block My Stuff: Fostering Personal Object Awareness in Multi-user Mixed Reality Environments
Talha Khan, David Lindlbauer
PACM HCI (ACM ISS 2024)

[C.5] Understanding Perceived Utility and Comfort of In-Home General-Purpose Sensing through Progressive Exposure
Pranut Jain, Andrew Xu, Thomas Downes, Injung Kim, **Talha Khan**, Jacob Biehl, Adam Lee
PACM HCI (ACM CSCW 2024)

[C.4] EndovasculAR: Utility of Mixed Reality to Segment Large Displays in Surgical Settings
Griffin Hurt, **Talha Khan**, Michael Kann, Edward Andrews, Jacob Biehl
IEEE VRW 2024

[C.3] Future of Surgical Mixed Reality: Cutting-Edge or Cutting Too Deep?
Talha Khan, Edward Andrews, Jacob Biehl
IEEE VRW 2024

[C.2] Understanding Effects of Visual Feedback Delay in AR on Fine Motor Surgical Tasks
Talha Khan, Toby Zhu, Thomas Downes, Lucille Cheng, Nicolás Kass, Edward Andrews, Jacob Biehl
IEEE TVCG (ISMAR 2023) **Best Paper Award**

[C.1] AR in the OR: exploring use of augmented reality to support endoscopic surgery
Talha Khan, Edward Andrews, Paul Gardner, Arka Mallela, Jeffrey Head, Joseph Maroon, Georgios Zenonos, Dmitriy Babichenko, Jacob Biehl
ACM IMX 2022

Journal Publications

[J.5] Application of Mixed Reality in Endoscopic Streaming during Skull Base Surgery
Yujin Choi, Shovan Bhatia, Adam Monek, Harrison Alexander, **Talha Khan**, Brendan Szczepkowski, Griffin Hurt, Anthony Tang, Michael Kann, Regan Shanahan, Brock Gjesdal, Rishi Basdeo, Kyle Affolter, David Fernandes Cabral, Georgios Zenonos, Paul Gardner, Jacob Biehl, Edward Andrews
Journal of Neurological Surgery Part B: Skull Base 2026 (Abstract)
Operative Neurosurgery 2026 (Full Paper)

[J.4] Evolution of the meta-neurosurgeon: A systematic review of the current technical capabilities, limitations, and applications of augmented reality in neurosurgery
Nikhil Sharma, Arka Mallela, **Talha Khan**, Stephen Paul Canton, Nicolás Kass, Fritz Steuer, Jacquelyn Jardini, Jacob Biehl, Edward Andrews
Surgical Neurology International 2024

[J.3] Solar-powered Parking Analytics System using Deep Reinforcement Learning
Yoones Rezai, **Talha Khan**, Stephen Lee, Daniel Mossé
ACM TOSN 2023

[J.2] A systematic comparison of the accuracy of monocular RGB tracking and LiDAR for neuronavigation
Talha Khan, Jacob Biehl, Edward Andrews, Dmitriy Babichenko
Healthcare Technology Letters 2022

[J.1] Streaming 2D-Endoscopic Video into an Augmented Reality Headset Display: A Feasibility Study
Edward Andrews, **Talha Khan**, Arka Mallela, Joseph Maroon, Jacob Biehl, David Fernandes-Cabral, Zachary Gersey, Hussam Abou-Al-Shaar, Paul Gardner, Georgios Zenonos
Journal of Neurological Surgery Part B: Skull Base 2022

Mentorship Experience

- Khadija Arif — Jan. 2026 - Current
  MS Human-Centered Computing, Grand Valley State University, USA
- Liam Adams — Aug. 2025 - Current
  MS Human-Centered Computing, Grand Valley State University, USA
- Abigail Zimmerman — Sep. 2024 - Jan. 2025
  BS Digital Narrative and Interactive Design, University of Pittsburgh, USA
- Bo-Chen Kuo — Jan. 2024 - Apr. 2024
  MS Computer Science, University of Pittsburgh, USA
- Elliott Finney — Jan. 2024 - Apr. 2024
  BS Computer Science, University of Pittsburgh, USA
- Griffin Hurt — May. 2023 - Apr. 2024
  BS Computer Science, University of Pittsburgh, USA
- Chloe Dahan — Jun. 2021 - Aug. 2021
  BS Digital Narrative and Interactive Design, University of Pittsburgh, USA
- Aiden Dorneich — Jun. 2021 - Aug. 2021
  Research Intern, Facet Lab, University of Pittsburgh, USA

Honors and Awards

- CS 50 Fellowship Award, Department of Computer Science, University of Pittsburgh — Apr. 2024
- Best Journal Paper Award, IEEE International Symposium on Mixed and Augmented Reality — Oct. 2023
- Honorable Mention Award, Link Foundation Modelling, Simulation & Training Fellowship — Jun. 2022

Media Appearances

- **Pitt To The Point** — Mar. 2024
  Revolutionizing Healthcare With Mixed Reality

Invited Talks

- **Guest Lecture, Arizona State University** — Feb. 2025
  Impact of Visual Delays in Mixed Reality on Surgical Performance
- **Dean's Spotlight Series, University of Pittsburgh** — Jan. 2023
  AR in the OR: An Interdisciplinary Journey Towards the Holographic Operating Room

Leadership Experience

- **Business Manager** — Mar. 2024 - APR. 2025
  Doctoral Guild at Pitt, University of Pittsburgh, USA

- **Founding Member** JAN. 2021 - CURRENT
  Surreality Lab, University of Pittsburgh, USA

REVIEWING SERVICE

- ACM International Conference on Interactive Surfaces and Spaces (ACM ISS 2026)
- Springer Nature Virtual Reality (2026)
- IEEE International Conference on Virtual Reality and 3D User Interfaces *(IEEE VR 2026)*
- ACM CHI Conference on Human Factors in Computing Systems *(ACM CHI 2026)*
- ACM Symposium on Spatial User Interaction *(ACM SUI 2025)*
- ACM CHI Conference on Human Factors in Computing Systems *(ACM CHI 2025)*
- ACM Symposium on User Interface Software and Technology *(ACM UIST 2024)*
- ACM CHI Conference on Human Factors in Computing Systems *(ACM CHI 2024)*
- ACM CHI Conference on Human Factors in Computing Systems *(ACM CHI 2023)*
- ACM Conference on Computer-Supported Cooperative Work and Social Computing *(ACM CSCW 2023)*

VOLUNTEER SERVICE

- Student Volunteer, ACM Conference on Interactive Surfaces and Spaces (ACM ISS 2023)
- Student Volunteer, ACM Conference on Human Factors in Computing Systems (ACM CHI 2022)

REFERENCES

DR. JACOB BIEHL
Associate Professor, Department of Computer Science, University of Pittsburgh, USA
https://jtbiehl.github.io/ | ✉ biehl@pitt.edu

DR. DAVID LINDLBAUER
Assistant Professor, Human-Computer Interaction Institute, Carnegie Mellon University, USA
https://www.davidlindlbauer.com/ | ✉ davidlindlbauer@cmu.edu

DR. ERIN WALKER
Professor, Department of Computer Science, University of Pittsburgh, USA
https://erinwalker.academicwebsite.com/ | ✉ eawalker@pitt.edu

DR. EDWARD ANDREWS
Assistant Professor, Department of Neurological Surgery, University of Pittsburgh, USA
✉ andrewse2@upmc.edu